Report on

Student Record Management System using Blockchain Technology



By

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I would also like to acknowledge Udemy for providing online courses related to web development and Solidity programming. These courses helped me understand the basic concepts required to develop a simple blockchain-based web application.

The knowledge acquired during the training was applied to the development of the Student Record Management System Using Blockchain Technology. The project provided practical experience in developing a website, working with smart contracts and storing and retrieving student academic information.

Finally, I would like to thank everyone who supported and encouraged me throughout the training and project development process.

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ABSTRACT

This industrial training project focuses on the development of a basic Student Record Management System Using Blockchain Technology. The project was developed by applying knowledge gained through online training courses in web development and Solidity programming.

The system provides a simple website through which student academic information can be entered, stored and retrieved. The application uses a Solidity smart contract to manage student records, while the frontend provides the interface for interacting with the system. MetaMask and Ethers.js are used to connect the website with the blockchain-based smart contract.

The student record contains details such as the student's name, roll number, registration number, department, semester, CGPA and wallet address. The system allows a record to be submitted through the website and subsequently retrieved using the relevant student identifier.

The main purpose of the project is to gain practical knowledge of web development and blockchain-based smart contract interaction. The project demonstrates a basic application of blockchain technology in student record management.

Keywords: Blockchain, Solidity, Smart Contract, Ethereum, Student Records, Web Development, MetaMask.

CHAPTER 1: INTRODUCTION

1.1 Background of Industrial Training

Industrial training provides students with an opportunity to apply the knowledge acquired through academic studies to practical projects. It helps students understand the development process, improve programming skills and gain experience with tools and technologies used in software development.

Computer science and engineering includes several technical areas, including web development, programming, blockchain and smart contract development. Learning these areas through practical work helps students understand how different technologies can be combined to develop a working application.

As part of the industrial training programme, two online courses were undertaken through Udemy. The courses focused on web development and Solidity programming. The knowledge gained from these courses was used to develop a basic Student Record Management System Using Blockchain Technology.

1.2 Introduction to the Project

The Student Record Management System Using Blockchain Technology is a basic website developed for storing and retrieving student academic records through a smart contract.

The website provides an interface where student information can be entered. The information is then submitted to the smart contract through blockchain interaction. Previously stored records can also be retrieved through the website.

The project combines a simple frontend with blockchain functionality. The frontend is developed using HTML, CSS and JavaScript, while Solidity is used for writing the smart contract.

The project is intended to demonstrate the basic working of a blockchain-based record management application.

1.3 Motivation

The motivation behind the project is to gain practical knowledge of web development and blockchain technology by implementing a simple working application.

Student records contain important academic information, including student identification details and academic performance. The project explores how a smart contract can be used to store and retrieve such information in a basic demonstration system.

The project also provides an opportunity to understand how a website can interact with a blockchain smart contract through a wallet and a JavaScript library.

1.4 Problem Statement

Student information needs to be stored and accessed in an organized manner. This project focuses on developing a basic website that allows student academic details to be submitted and retrieved using blockchain-based smart contract functionality.

The problem addressed by the project is:

To develop a simple web-based student record management system that stores and retrieves student academic information using a Solidity smart contract.

1.5 Objectives

The main objectives of the project are:

- To understand the basic concepts of web development.
- To learn the fundamentals of Solidity programming.
- To develop a simple student record management website.
- To create a smart contract for storing student information.
- To connect the website with the smart contract.
- To store student academic details through the website.
- To retrieve previously stored student records.
- To gain practical experience in blockchain application development.

1.6 Scope of the Project

The scope of the project is limited to developing a basic student record management website.

The system includes:

- A simple website interface.
- A student information form.
- Storage of student academic details through a smart contract.
- Retrieval of stored student records.
- Basic MetaMask wallet interaction.
- Blockchain communication through Ethers.js.

The project does not include advanced institutional management features, complex authentication systems or a complete university-level student database.

CHAPTER 2: INDUSTRIAL TRAINING OVERVIEW

2.1 Introduction

The industrial training was completed through online courses provided by Udemy. The training consisted of two major areas: web development and Solidity programming.

The knowledge gained from these courses was used to develop the Student Record Management System Using Blockchain Technology.

The training provided an introduction to the technologies required for developing a simple website and connecting it with a blockchain smart contract.

2.2 Training Details

Particular	Details
Training provider	Udemy
Training type	Online course-based training
Training duration	2 months
Total training hours	40 hours
Number of courses	Two
Course 1	The Complete Solidity Course – Blockchain – Zero to Expert
Course 2	Ultimate Web Development Course 2026 – Build Modern Websites
Project	Student Record Management System Using Blockchain Technology

2.3 Web Development Training

The web development training introduced the basic concepts required for creating a website.

The knowledge gained from this training was applied to the development of the frontend of the project. The frontend provides the structure and interface through which users can enter student information and view stored records.

The major learning areas included:

- HTML for creating webpage structure.
- CSS for styling the website.
- JavaScript for adding functionality.
- Creating forms.
- Handling user input.
- Displaying information on a webpage.
- Developing basic website interaction.

The web development knowledge helped in creating the user interface of the Student Record Management System.

2.4 Solidity Programming Training

The Solidity programming training introduced the basic concepts of smart contracts and blockchain application development.

Solidity is used to write smart contracts for Ethereum-compatible blockchain networks. The knowledge acquired from the training was applied to create a contract for storing and retrieving student information.

The learning areas included:

- Basic Solidity syntax.
- Contract structure.
- Functions.
- Data types.
- Variables.
- Smart contract interaction.
- Storing data in a contract.

- Retrieving data from a contract.

The Solidity training helped in understanding the blockchain component of the project.

2.5 Technologies Used

Technology	Purpose
HTML	Creating the website structure
CSS	Styling and designing the website
JavaScript	Implementing website functionality
Solidity	Writing the smart contract
Ethereum	Blockchain platform
MetaMask	Connecting the wallet
Ethers.js	Connecting the website with the smart contract
Remix IDE	Developing and testing the smart contract

2.6 Learning Outcomes

The industrial training helped in developing the following skills:

- Basic website development.
- Frontend design and implementation.
- JavaScript-based interaction.
- Solidity programming.
- Understanding smart contracts.
- Connecting a website with a blockchain.
- Storing and retrieving data using contract functions.
- Debugging and testing a working application.

CHAPTER 3: BASIC TECHNICAL BACKGROUND

3.1 Introduction

This chapter explains the basic technologies used during the development of the Student Record Management System.

The project combines web development technologies with blockchain and smart contract functionality. Understanding the role of each technology is important for understanding the working of the application.

3.2 Overview of Web Development

Web development refers to the process of creating websites and web applications.

A website generally consists of a frontend that users interact with. HTML, CSS and JavaScript are commonly used to develop the frontend.

- HTML: Used to create the structure of a webpage.
- CSS: Used to control the appearance, layout and styling.
- JavaScript: Used to implement interaction and functionality.

In this project, these technologies are used to create the student record management interface.

3.3 Introduction to Blockchain

Blockchain is a technology used to maintain records of transactions or information through a distributed ledger.

Information is recorded through transactions, and the network maintains the state of the blockchain. Blockchain technology is commonly associated with cryptocurrencies and smart contract applications.

In this project, blockchain is used as the platform on which the smart contract operates.

3.4 Introduction to Ethereum

Ethereum is a blockchain platform that supports smart contracts and decentralized applications.

It allows developers to write and deploy programs known as smart contracts. These contracts can execute predefined functions and maintain data on the blockchain.

The project uses Ethereum-compatible smart contract functionality to store and retrieve student records.

3.5 Smart Contracts

A smart contract is a program deployed on a blockchain network. It contains functions that can be executed according to the logic defined in the contract.

In this project, the smart contract is used to manage student records. It provides functions for adding student information and retrieving previously stored records.

The smart contract acts as the main component responsible for managing the student record data.

3.6 Solidity Programming

Solidity is a programming language used for developing smart contracts on Ethereum-compatible blockchain networks.

It supports functions, variables, data types and other programming concepts required for defining smart contract logic.

In this project, Solidity is used to define the structure and functions required for storing and retrieving student academic details.

3.7 MetaMask and Ethers.js

MetaMask is a wallet that allows users to connect a website with a blockchain network and authorize transactions.

Ethers.js is a JavaScript library used to communicate with Ethereum-compatible blockchain networks and smart contracts.

In the project:

- MetaMask is used for wallet connection.
- Ethers.js is used for communication between the website and smart contract.
- The smart contract handles the student record operations.

CHAPTER 4: PROJECT ANALYSIS AND REQUIREMENTS

4.1 Introduction

This chapter describes the problem addressed by the project and the basic requirements of the Student Record Management System.

The system is designed to provide a simple interface for entering, storing and retrieving student academic information.

4.2 Problem Analysis

Student records contain important information such as student names, roll numbers, registration numbers and academic performance.

The objective of this project is to create a simple system that demonstrates how student information can be stored and retrieved using a smart contract.

The project focuses on the basic functionality of record management rather than implementing a complete institutional student information system.

4.3 Proposed System

The proposed system is a simple website connected to a Solidity smart contract.

The user enters student information through the website. The application communicates with the smart contract to store the information. The stored information can later be retrieved using the relevant student identifier.

The main operations of the proposed system are:

1. Open the website.
2. Connect the wallet.
3. Enter student academic details.
4. Submit the record.

5. Store the record through the smart contract.
6. Retrieve the stored record.
7. Display the record on the website.

4.4 Functional Requirements

Requirement	Description
Website interface	The system should provide a basic user interface.
Wallet connection	The system should allow the user to connect MetaMask.
Student form	The system should provide fields for student details.
Record storage	The system should store student information using the smart contract.
Record retrieval	The system should retrieve stored student information.
Data display	The system should display retrieved student details.
Status messages	The system should provide relevant feedback to the user.

4.5 Non-Functional Requirements

4.5.1 Usability

The website should be simple and easy to understand.

4.5.2 Functionality

The main record storage and retrieval operations should work correctly.

4.5.3 Compatibility

The website should be accessed through a browser that supports the required wallet functionality.

4.5.4 Maintainability

The code should be organized so that basic changes and debugging can be performed when required.

4.6 Hardware and Software Requirements

Hardware Requirements

- Laptop or desktop computer.
- Stable internet connection.
- Sufficient system resources to run the browser and development tools.

Software Requirements

- Web browser.
- Remix IDE.
- MetaMask.
- HTML, CSS and JavaScript environment.
- Ethers.js library.
- Solidity compiler.

CHAPTER 5: SYSTEM DESIGN AND METHODOLOGY

5.1 Introduction

The system design explains how the main components of the Student Record Management System work together.

The project consists of a frontend website, JavaScript functionality, MetaMask, Ethers.js and a Solidity smart contract.

5.2 System Overview

The website provides the interface through which users can enter and retrieve student records. The frontend communicates with the smart contract through Ethers.js. MetaMask is used to connect the user's wallet and authorize the required blockchain transaction.

The smart contract manages the storage and retrieval of student information.

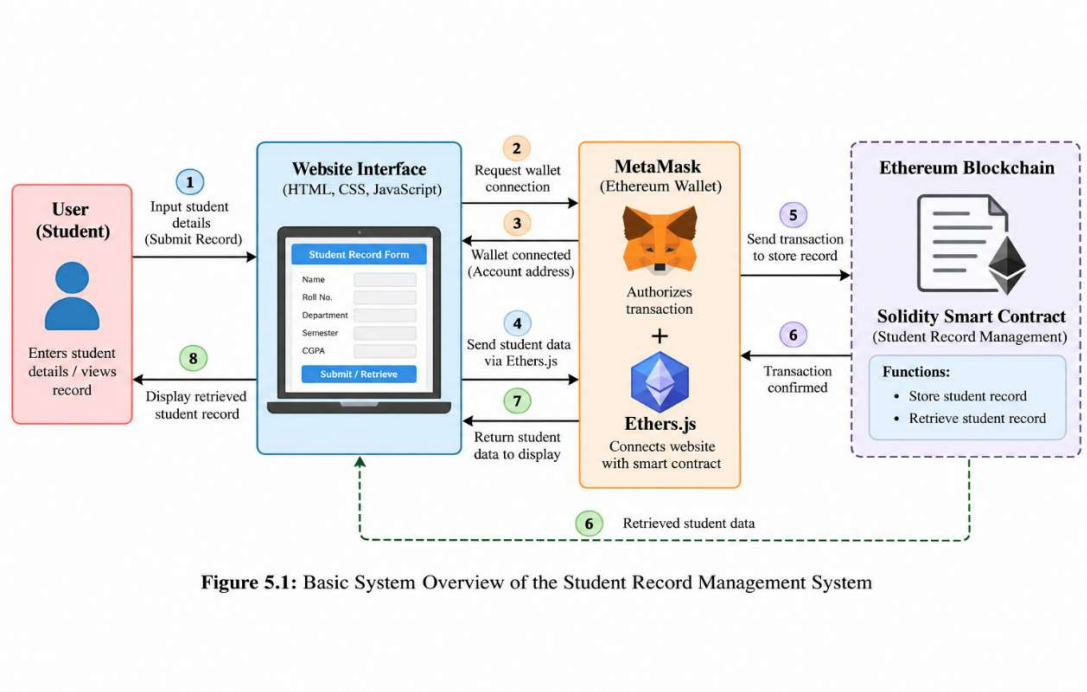


Figure 5.1: Basic System Overview of the Student Record Management System

Figure 5.1: Basic System Overview

5.3 System Components

5.3.1 Website Interface

The website provides the interface for entering and displaying student information.

5.3.2 JavaScript Functionality

JavaScript handles user interaction, form processing and communication with the smart contract.

5.3.3 MetaMask

MetaMask connects the user's wallet with the website and allows transactions to be authorized.

5.3.4 Smart Contract

The smart contract manages the functions for storing and retrieving student records.

5.4 Working Methodology

The project was developed through the following steps:

1. Learning the basic concepts of web development.
2. Learning Solidity programming and smart contract concepts.
3. Creating the website interface.
4. Developing the smart contract.
5. Connecting the website with the smart contract.
6. Testing the storage and retrieval functionality.
7. Reviewing the working website.

5.5 Student Record Storage Process

The student record storage process follows these steps:

1. The user opens the student record form.
2. The user enters the student's academic details.
3. The application checks the entered information.
4. The application communicates with the smart contract.
5. The user authorizes the transaction through MetaMask.
6. The smart contract stores the student record.
7. The application displays the relevant status or result.

5.6 Student Record Retrieval Process

The retrieval process follows these steps:

1. The user opens the record retrieval section.
2. The user enters the required student identifier.
3. The application calls the smart contract retrieval function.
4. The smart contract returns the stored record.
5. The website displays the student information.

The system is designed to demonstrate basic student record storage and retrieval rather than advanced academic administration.

CHAPTER 6: PROJECT IMPLEMENTATION

6.1 Introduction

The implementation phase involved creating the website interface and connecting it with the Solidity smart contract.

The project was developed using basic web development technologies and blockchain interaction tools.

6.2 Website Interface

The website interface was developed using HTML and CSS.

HTML was used to create the structure of the website, while CSS was used to style the page and organize the different sections.

The website contains a basic interface for:

- Introducing the project.
- Connecting the wallet.
- Entering student information.
- Displaying stored records.
- Retrieving student records.



Figure 6.1: Website Interface

6.3 Student Record Form

The student record form allows the user to enter academic details.

The information included in the form consists of:

- Student name.
- Roll number.
- Registration number.
- Department.
- Semester.
- CGPA.
- Wallet address.

The form provides an interface through which the user can submit student information to the smart contract.

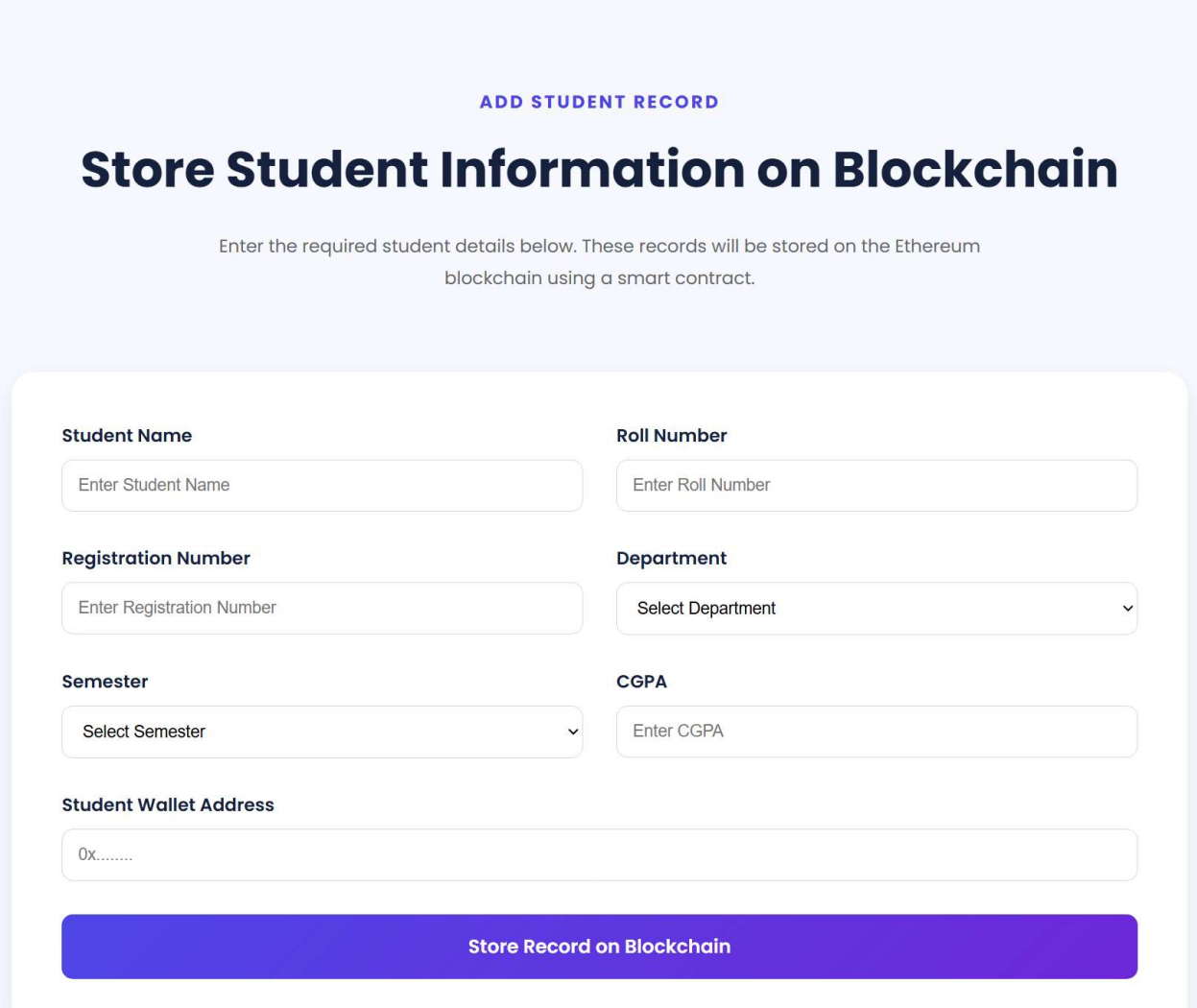


Figure 6.2: Student Record Input Form

6.4 Smart Contract Implementation

The smart contract was written using Solidity.

The purpose of the contract is to store and retrieve student academic information. The contract includes functions for adding student records and retrieving records using the relevant student identifier.

The contract stores the information provided by the website through the blockchain transaction process.

The smart contract was developed using Solidity to store and retrieve student records. It includes a function for adding student information and a function for retrieving records using the student roll number. The stored information includes the student's name, roll number, registration number, department, semester, CGPA, and wallet address. The contract functions are accessed through the website using Ethers.js.

6.5 Blockchain Interaction

The website communicates with the smart contract using Ethers.js.

When the user submits a record, the JavaScript code collects the information from the form and calls the relevant smart contract function.

MetaMask is used to authorize the transaction. After the transaction is completed, the application provides feedback to the user.

This integration demonstrates how a website can interact with a blockchain-based smart contract.

6.6 Record Retrieval

The record retrieval functionality allows the user to obtain previously stored student information.

The website sends a request to the smart contract using the relevant student identifier. The returned information is then displayed on the website.

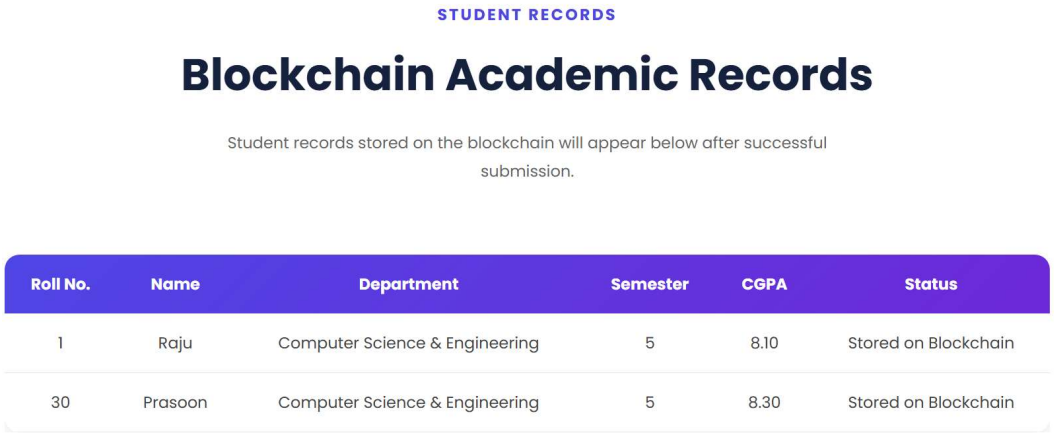


Figure 6.3: Retrieved Student Record

CHAPTER 7: TESTING AND RESULTS

7.1 Introduction

Testing was performed to verify the basic functionality of the Student Record Management System.

The main focus of testing was to check whether the website could connect to MetaMask, store student records and retrieve previously stored records.

7.2 Testing Methodology

The application was tested by performing the main operations available on the website.

The following activities were checked:

- Opening the website.
- Connecting MetaMask.
- Entering student information.
- Submitting a student record.
- Checking the storage result.
- Retrieving a student record.
- Displaying the retrieved information.

7.3 Test Cases

Test ID	Test Case	Expected Result	Status
TC01	Open the website	Website opens successfully	[Pass]
TC02	Connect MetaMask	Wallet connects successfully	[Pass]
TC03	Enter student details	Details are accepted by the form	[Pass]
TC04	Submit student record	Record submission is initiated	[Pass]
TC05	Store student record	Record is stored through the smart contract	[Pass]
TC06	Retrieve existing record	Correct record is displayed	[Pass]
TC07	Display academic details	Student details and CGPA are displayed	[Pass]

Update the status column based on your actual testing.

7.4 Testing Results

The testing process focused on the main functionality of the website.

The wallet connection was tested to verify that the website could interact with MetaMask. The record storage functionality was checked by submitting student information through the form. The retrieval functionality was tested by requesting a previously stored student record.

CHAPTER 8: LEARNING OUTCOMES AND LIMITATIONS

8.1 Skills Acquired

The project helped in developing the following technical skills:

Web Development Skills

- Creating a basic website using HTML.
- Styling webpages using CSS.
- Implementing website functionality using JavaScript.
- Creating and handling forms.

Blockchain Skills

- Understanding blockchain basics.
- Learning Solidity programming.
- Understanding smart contract functionality.
- Connecting a website with a smart contract.
- Using MetaMask for blockchain interaction.
- Using Ethers.js for communication with the contract.

Practical Development Skills

- Debugging code.
- Testing website functionality.
- Understanding the relationship between frontend and smart contract components.
- Applying course-based knowledge to a practical project.

8.2 Challenges Encountered

During the development of the Student Record Management System, some challenges were encountered while connecting the website with MetaMask and the Solidity smart contract. Configuring the smart contract address and ABI correctly was necessary for proper communication between the website and the blockchain. Understanding the transaction process and retrieving stored student records also required debugging and testing. Additionally, ensuring that the website was connected to the correct blockchain network was an important part of the development process.

8.3 Solutions Implemented

The encountered issues were addressed by reviewing the relevant code, checking the connection between the website and smart contract and testing the application repeatedly.

The challenges were addressed by reviewing the JavaScript code, checking the smart contract address and ABI, and verifying the MetaMask network connection. The website and smart contract were tested repeatedly to identify and resolve errors. The storage and retrieval functions were checked to ensure that student records could be submitted and displayed correctly. This process helped improve the understanding of blockchain interaction and practical debugging.

The development process helped in understanding the importance of debugging and verifying each component of the application.

8.4 Project Limitations

The project has the following limitations:

1. It is a basic demonstration website.
2. It does not provide a complete university-level student management system.
3. It does not include advanced user authentication.
4. It has limited record management functionality.
5. It is intended for educational and practical learning purposes.
6. It does not include advanced privacy or institutional database integration.

These limitations are consistent with the basic scope of the project.

8.5 Future Scope

The project can be improved in the future by adding additional features such as:

- Student record updating.
- Student record deletion, where appropriate.
- Improved form validation.
- Administrative access.
- Better record search functionality.
- Additional user interface improvements.
- Integration with a more complete student management system.

These features are possible future improvements and are not part of the current implementation unless they have already been developed.

CHAPTER 9: CONCLUSION

9.1 Summary

The industrial training provided practical knowledge in web development and Solidity programming. The knowledge gained from the training was applied to develop a basic Student Record Management System Using Blockchain Technology.

The project combines a simple website interface with a Solidity smart contract. The website allows student academic details to be entered, stored and retrieved through blockchain interaction.

The project provided practical experience in creating a website, handling form information, connecting MetaMask and interacting with a smart contract through Ethers.js.

9.2 Conclusion

The Student Record Management System Using Blockchain Technology was developed as a basic practical project during the industrial training programme.

The project demonstrates how web development technologies can be combined with Solidity smart contracts to create a simple student record management application. The system allows student academic information to be submitted and retrieved through the website.

The training helped improve understanding of HTML, CSS, JavaScript, Solidity, smart contracts and blockchain interaction. It also provided experience in debugging, testing and applying theoretical knowledge to a practical implementation.

Although the project is basic in nature, it provides a foundation for further learning and development in web development and blockchain-based applications.

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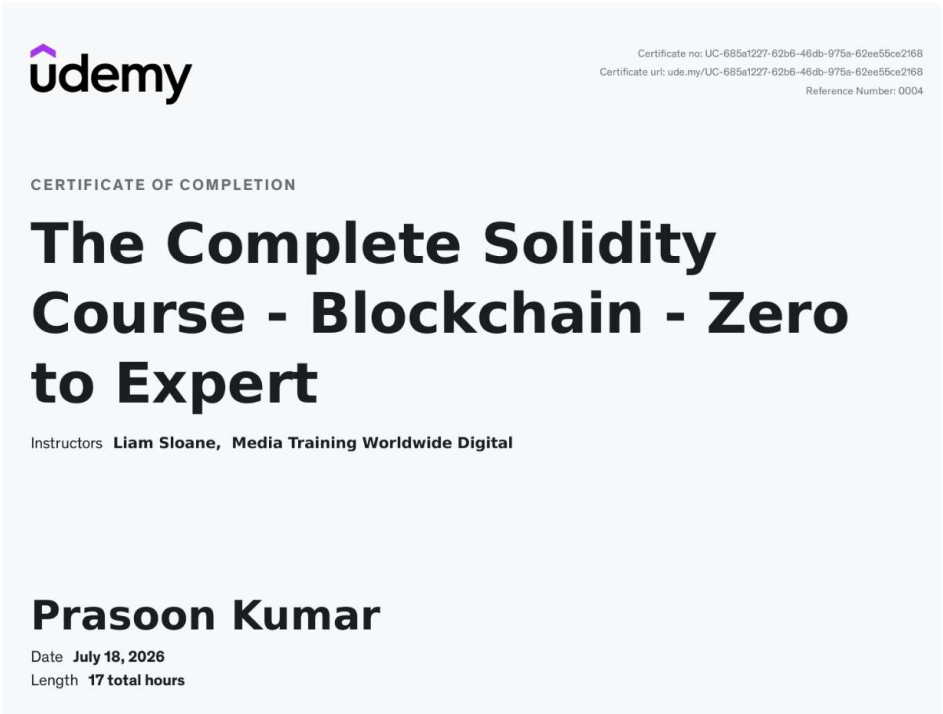
CERTIFICATE OF COMPLETION

Ultimate Web Development Course 2026 - Build Modern Websites

Instructors **Haris Ali Khan**

Prasoon Kumar

Date **Sept. 9, 2026**
Length **22 total hours**



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Reference Number: 0004

CERTIFICATE OF COMPLETION

The Complete Solidity Course - Blockchain - Zero to Expert

Instructors **Liam Sloane, Media Training Worldwide Digital**

Prasoon Kumar

Date **July 18, 2026**
Length **17 total hours**